

5. Project Development Phase

Project Title

Script-Controlled ACL – Restrict Record Access Based on Field Value

Development Overview

The Project Development Phase focuses on implementing the Script-Controlled ACL based on the design and requirements identified in the previous phases. The main purpose is to restrict access to records according to a specific field value.

The ACL is configured in the ServiceNow platform and a script is used to evaluate the required condition.

Table and Field Selection

First, the required table is selected for implementing the access control. A specific field from the table is selected for determining record access.

For example, the Department field can be used. Each record contains a department value, which is checked by the ACL script.

ACL Configuration

An Access Control List is created for the selected table and required operation. The ACL defines the rule that determines whether a user can access a record.

The required script is added to the ACL to perform the field-value check.

Script Implementation

The ACL script checks the selected field value and compares it with the required condition.

The basic logic is:

Field Value Matches → Access Allowed

Field Value Does Not Match → Access Restricted

This allows the system to make an access decision for each individual record.

Development Steps

The main development steps are:

1. Select the required table.
2. Select the field used for access control.
3. Create sample records with different field values.
4. Create the required ACL.
5. Add the field-value checking script.
6. Save and activate the ACL.
7. Test the configuration with different users.
8. Access Control Example

Suppose a record has Department = Computer Science.

If the user's department is also Computer Science, the ACL allows access.

If the user's department is different, the ACL restricts access.

This demonstrates record-level access control based on a field value.

Error Correction

During development, errors may occur because of incorrect field names, script conditions, or ACL configuration. These errors are checked and corrected before the final testing phase.

The ACL is reviewed to make sure that authorized users receive access and unauthorized users are restricted.

Expected Result

After development, the system should have a working Script-Controlled ACL. The ACL should automatically evaluate the selected field value when a user requests a record.

The expected result is:

Matching condition → Access Allowed

Non-matching condition → Access Restricted

Conclusion

The Project Development Phase implements the Script-Controlled ACL and its field-value-based access logic. The table, field, ACL, and script are configured to control record access.

After successful development, the system is ready for the Project Testing Phase.